SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – EXPERIMENTATION

Course Code:	ITA03	**Course Title:**	Mobile computing
CO Assessed:	CO2	**Assessment:**	Assessment Tool 2 – EXPERIMENTATION
Weightage:	30 % of CIA	**Total Marks:**	100 marks
CO2 Statement:	*Compare mobile communication technologies, including multiple access techniques and network evolution, based on performance and applications.*		
Student Name:	K Sumanth Kumar Reddy	**Reg. No.:**	192425222
Section / Batch:	AIML	**Date:**	24-10-2026

Code of Conduct

I K Sumanth Kumar Reddy(192425222) *(Name / Reg. No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: K Sumanth

Instructions

- This test contains FIVE questions.
- Duration: 60 minutes (homework)

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Annexure A - QUESTIONS

1. Figma-Based Mobile IP Communication Workflow Analysis (BL4 – Analyze)

Design an interactive Figma prototype illustrating the complete workflow of Mobile IP communication when a Mobile Node moves from its Home Network to a Foreign Network. Your design should include the Home Agent (HA), Foreign Agent (FA), Correspondent Node (CN), Care-of Address (CoA), Registration Process, and Packet Tunneling.

Analyze the communication process by identifying:

- The purpose of each network component.
- Points where packet delay or packet loss may occur.
- The effect of node mobility on communication performance.
- Possible improvements to reduce latency.

Deliverables:

- Figma (.fig) design
- GitHub repository containing exported images/PDF and documentation (README).

2. Comparative Network Architecture Design Using Figma (BL4 – Analyze)

Using Figma or draw.io, create a comparative architectural diagram showing both a Cellular Wireless Network and a Mobile Ad Hoc Network (MANET) operating in the same smart city environment.

Analyze the architectures by comparing:

- Infrastructure dependency.
- Routing mechanisms.
- Scalability.
- Fault tolerance.
- Mobility support.
- Suitable application scenarios.

Highlight the advantages and limitations of each architecture through annotations in your design.

Deliverables:

- Editable design file.
- GitHub repository with screenshots, comparison report, and design assets.

3. Routing Protocol Visualization and Analysis (BL4 – Analyze)

Develop an interactive visualization using Figma, Mermaid.js, D3.js, or any visualization framework illustrating the operation of one Proactive Routing Protocol and one Reactive Routing Protocol in a mobile wireless network.

Analyze:

- Route discovery process.
- Route maintenance mechanism.
- Routing overhead.
- Delay in establishing communication.
- Performance under high node mobility.

Include a comparative summary explaining which protocol performs better under different mobility conditions.

Deliverables:

- Source code/design files.
- GitHub repository with implementation, screenshots, and analysis report.

4. Mobile IP Application Design for Smart Healthcare (BL4 – Analyze)

Design a Smart Healthcare Mobile IP system using Figma that enables doctors, ambulances, hospitals, and patients to remain connected while moving across different wireless networks.
Analyze:

- How Mobile IP maintains uninterrupted communication.
- Challenges during handoff between networks.
- Security vulnerabilities during mobility.
- Possible design improvements to increase reliability.

Provide navigation between screens demonstrating different mobility scenarios.
Deliverables:

- Interactive prototype.
- GitHub repository containing design files, exported prototype, and documentation.

5. Wireless Network Decision Dashboard Using GitHub Project (BL4 – Analyze)

Create a web-based dashboard (using HTML/CSS/JavaScript, React, or another programming framework) or an interactive Figma dashboard that helps network engineers determine whether a given scenario should use Mobile IP, Cellular Networks, or Ad Hoc Networks.
The dashboard should analyze multiple parameters such as:

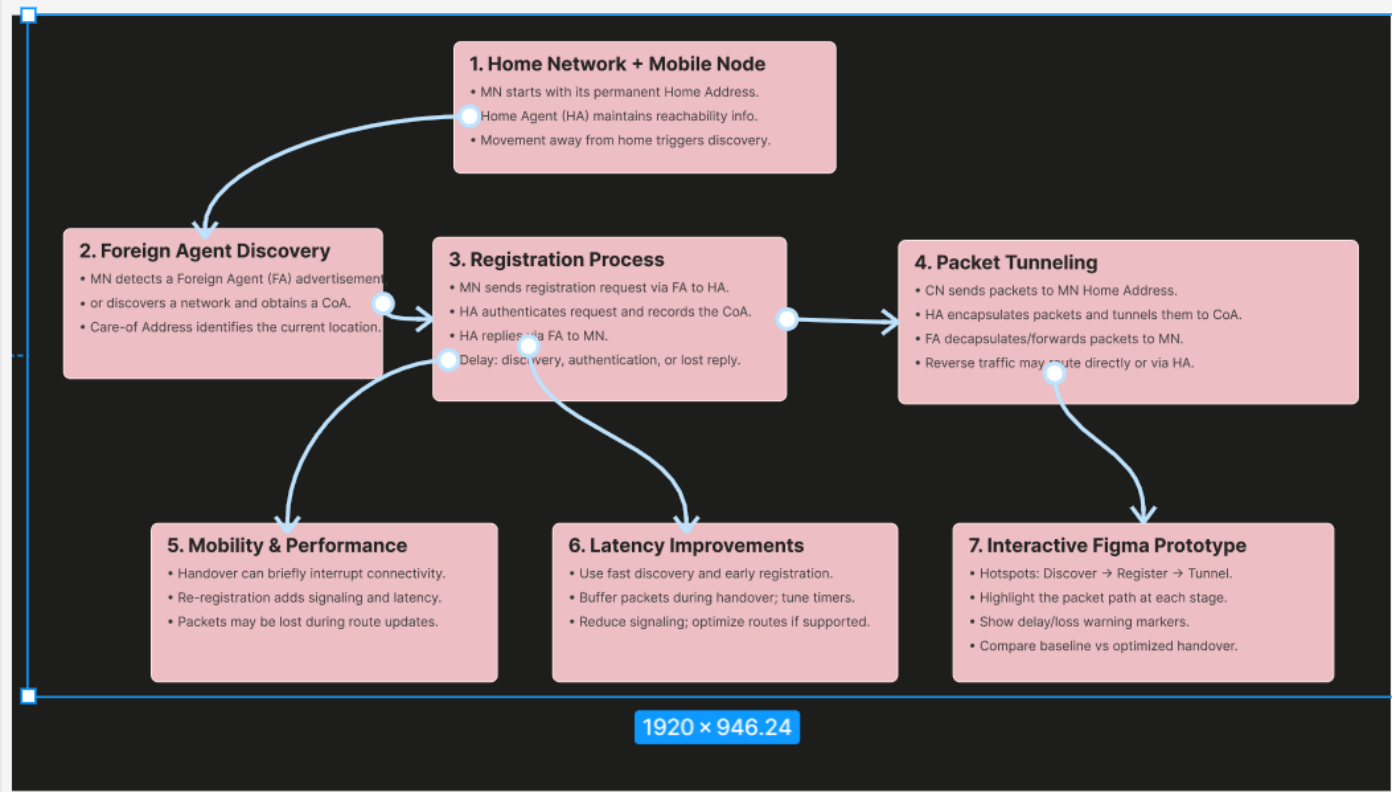
- Node mobility.
- Infrastructure availability.
- Network size.
- Routing complexity.
- Energy consumption.
- Reliability requirements.

The system should justify its recommendation based on the analyzed parameters.
Deliverables:

- Source code or Figma project.
- GitHub repository with complete code/design, README, screenshots, and analysis report.

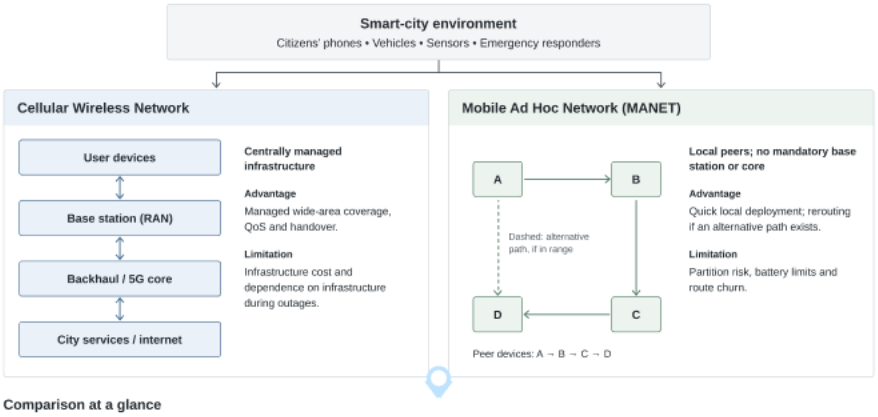
Criterion	Marks
Design Quality (Figma/Programming Tool)	20
Technical Analysis	30
Correctness of Mobile Networking Concepts	20
GitHub Repository Organization & Documentation	15
Creativity and Visualization	15
Total	**100 Marks**

1.


1. Home Network + Mobile Node
• MN starts with its permanent Home Address.
• Home Agent (HA) maintains reachability info.
• Movement away from home triggers discovery.
2. Foreign Agent Discovery
• MN detects a Foreign Agent (FA) advertisement
• or discovers a network and obtains a CoA.
• Care-of Address identifies the current location.
3. Registration Process
• MN sends registration request via FA to HA.
• HA authenticates request and records the CoA.
• HA replies via FA to MN.
• Delay: discovery, authentication, or lost reply.
4. Packet Tunneling
• CN sends packets to MN Home Address.
• HA encapsulates packets and tunnels them to CoA.
• FA decapsulates/forwards packets to MN.
• Reverse traffic may route directly or via HA.
5. Mobility & Performance
• Handover can briefly interrupt connectivity.
• Re-registration adds signaling and latency.
• Packets may be lost during route updates.
6. Latency Improvements
• Use fast discovery and early registration.
• Buffer packets during handover; tune timers.
• Reduce signaling; optimize routes if supported.
7. Interactive Figma Prototype
• Hotspots: Discover → Register → Tunnel.
• Highlight the packet path at each stage.
• Show delay/loss warning markers.
• Compare baseline vs optimized handover.
1920 × 946.24

2.

Cellular Wireless Network vs. Mobile Ad Hoc Network (MANET)



Comparison at a glance

Dimension	Cellular	MANET
Infrastructure	Requires RAN, backhaul and core.	Local peers only.
Routing	Operator-managed transport and core.	Distributed multihop: AODV (reactive) or OLSR (proactive).
Scalability	Planned cell / capacity expansion; constrained by spectrum and cost.	Contention and routing overhead grow; depends on density and protocol.
Fault tolerance	Engineered redundancy; infrastructure outages can affect service.	Reroutes only if connected alternatives remain.
Mobility	Coordinated handover between cells.	Dynamic route updates may cause delay or loss.
Applications	Citywide broadband and transit telemetry.	Disaster response, temporary field teams and local cooperative vehicles.

Key takeaway
Cellular: managed citywide service. MANET: infrastructure-free local coordination. Hybrid use is possible.

Conceptual diagram; coverage and connectivity depend on deployment.
Sources: 3GPP TS 23.501; RFC 2501; RFC 3561; RFC 3626.